

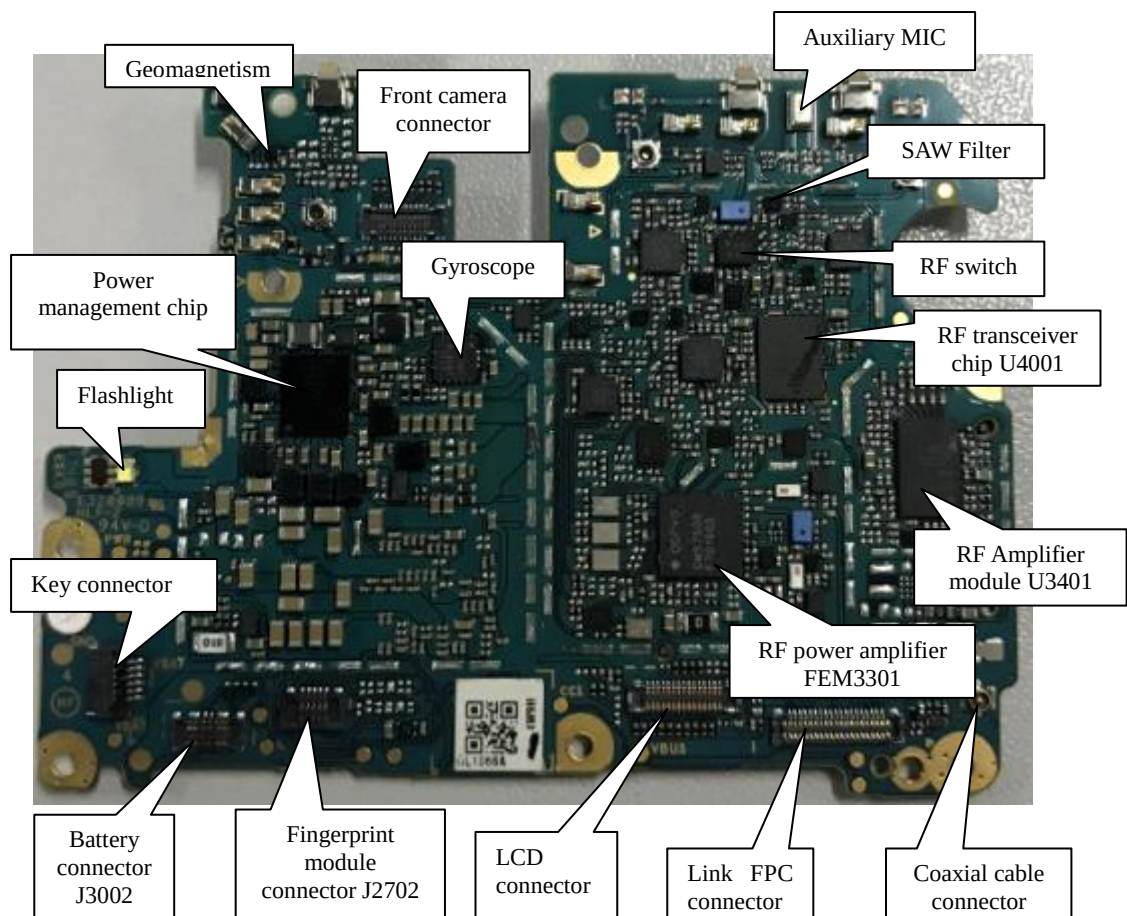
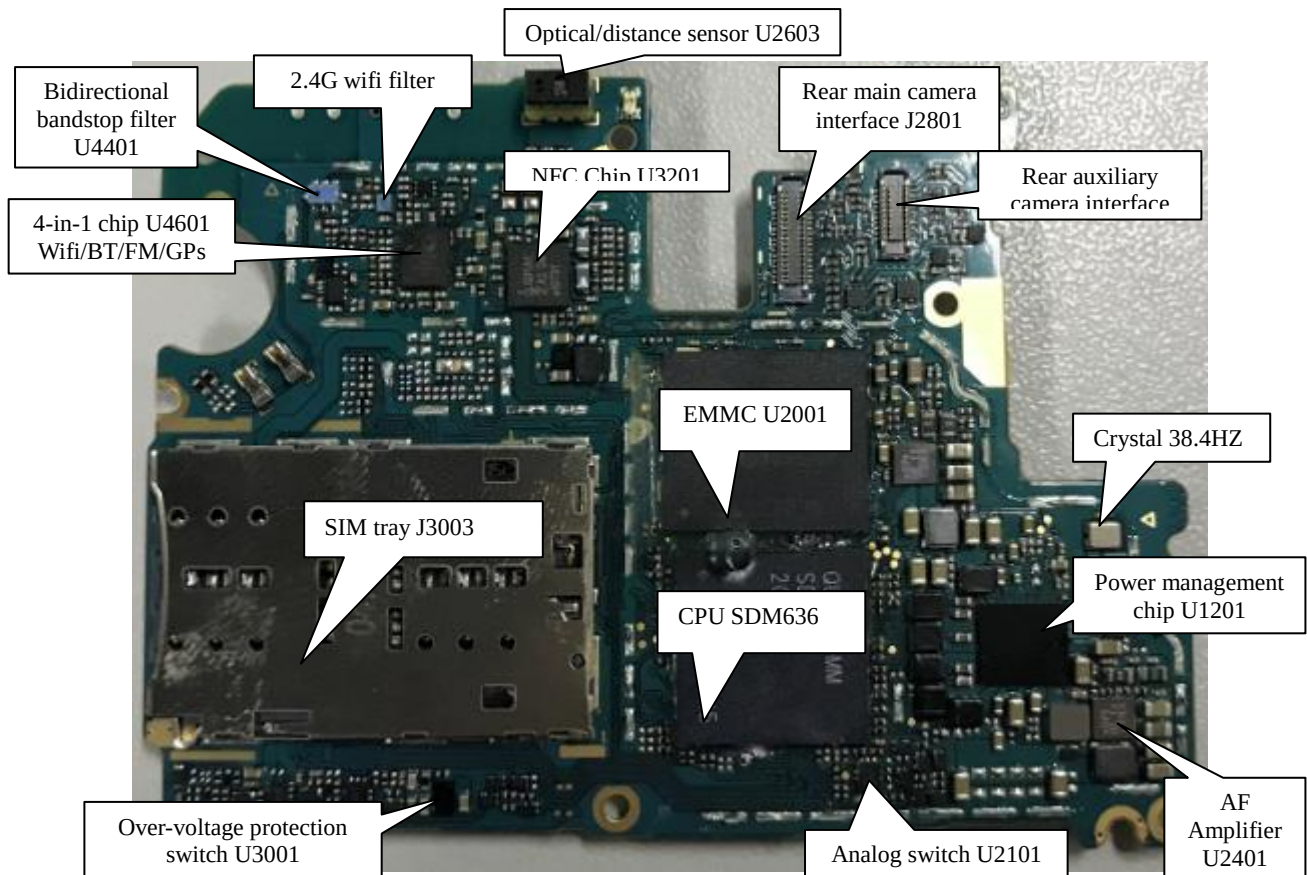
# ZE620KL L3-L4 Board Level Trouble Shooting Guide V1.0

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## 1. MB illustration

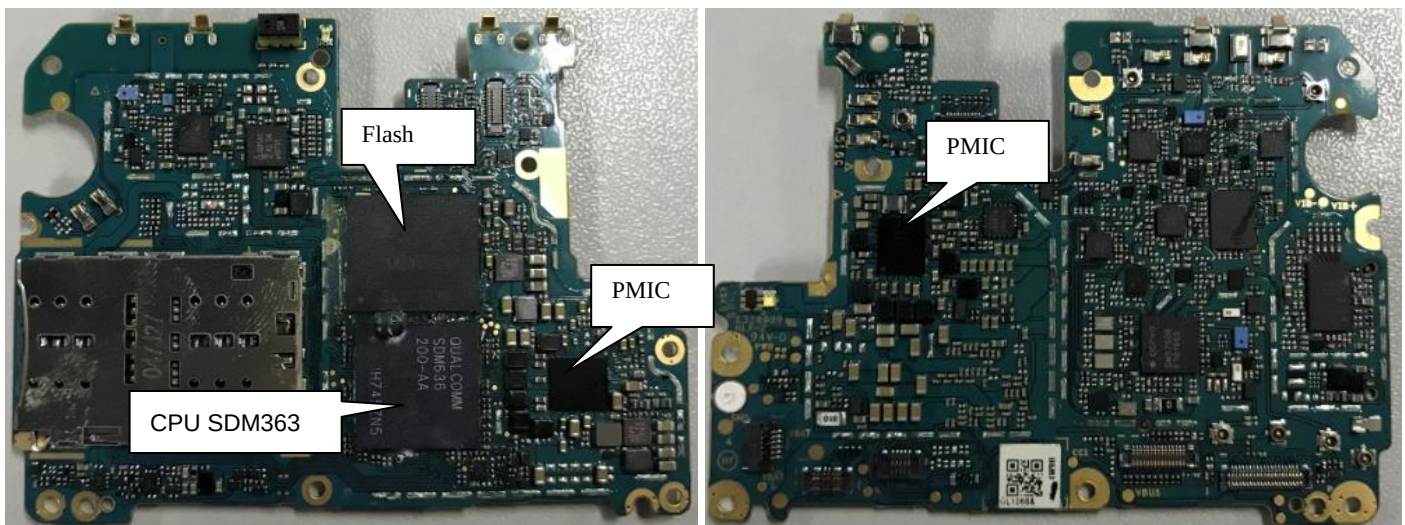


## 2. Power on failure

### Solution:

- 1 Check battery connector soldering and its body for oxidation or damage;
- 2 Check the following voltage for short circuits with GND or for any errors;

|          |            |
|----------|------------|
| VBAT     | 3.8--4.2V  |
| VCORE    | 0.7--1.4V  |
| VSYS     | 1.825/2.2V |
| VPA      | 0.5-3.4V   |
| VTXCO    | 2.8V       |
| VA       | 2.8V       |
| VCAMA    | 2.8V       |
| VCN28    | 2.8V       |
| VCAMA    | 2.8V       |
| VUSB     | 3.3V       |
| VEMC_3V3 | 3.3V       |
- 3 Check power on circuit for any errors;
- 4 For power on current around 20MA, check crystal for soldering errors;
- 5 For power on current errors above 1A, check VBAT circuit for short circuits to ground, troubleshoot ground diode at individual locations;
- 6 For power on current errors around 100MA, check for hot point on MB and troubleshoot;
- 7 Check soldering of each main chip with X-ray, CPU/Flash/PMIC;
- 8 Add-solder or re-solder main chip, validate soldering conditions;
- 9 Replace CPU/Flash/PMIC





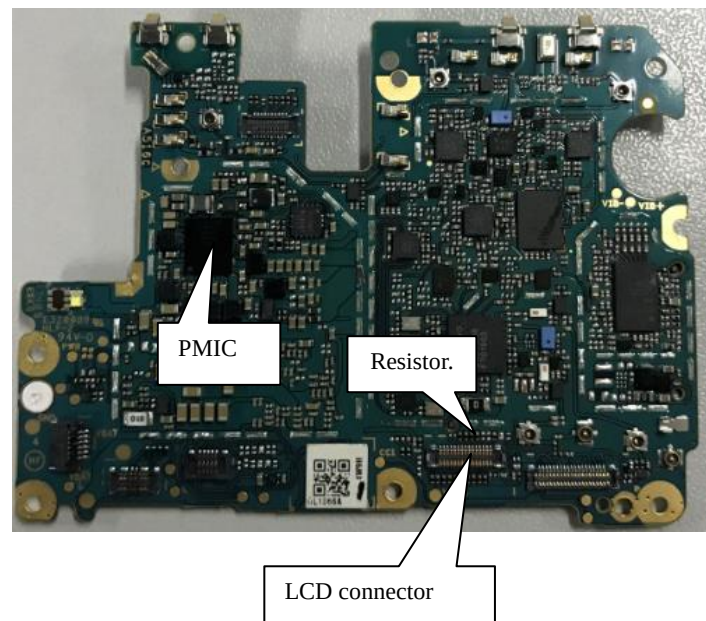
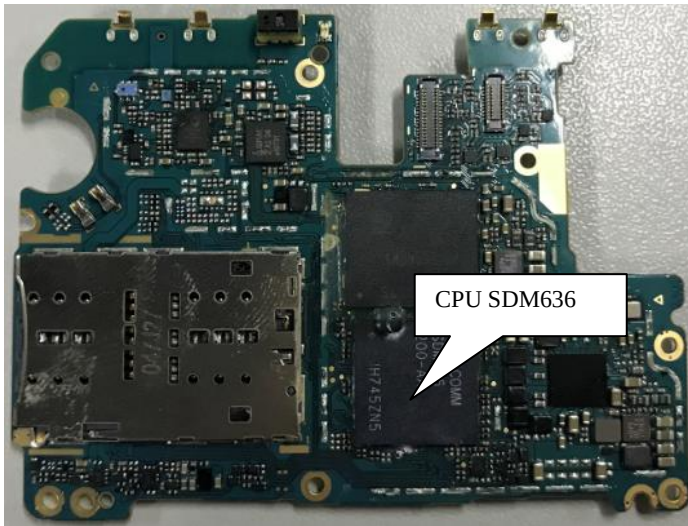
- 1 Check LCD connector for installation, re-install;
- 2 Check R1904 and R1905 for normal conduction, compare its voltage against that of R1904 and R1905 on normal MB;
- 3 Check VBAT for normal power supply;

#### 4. Display failure

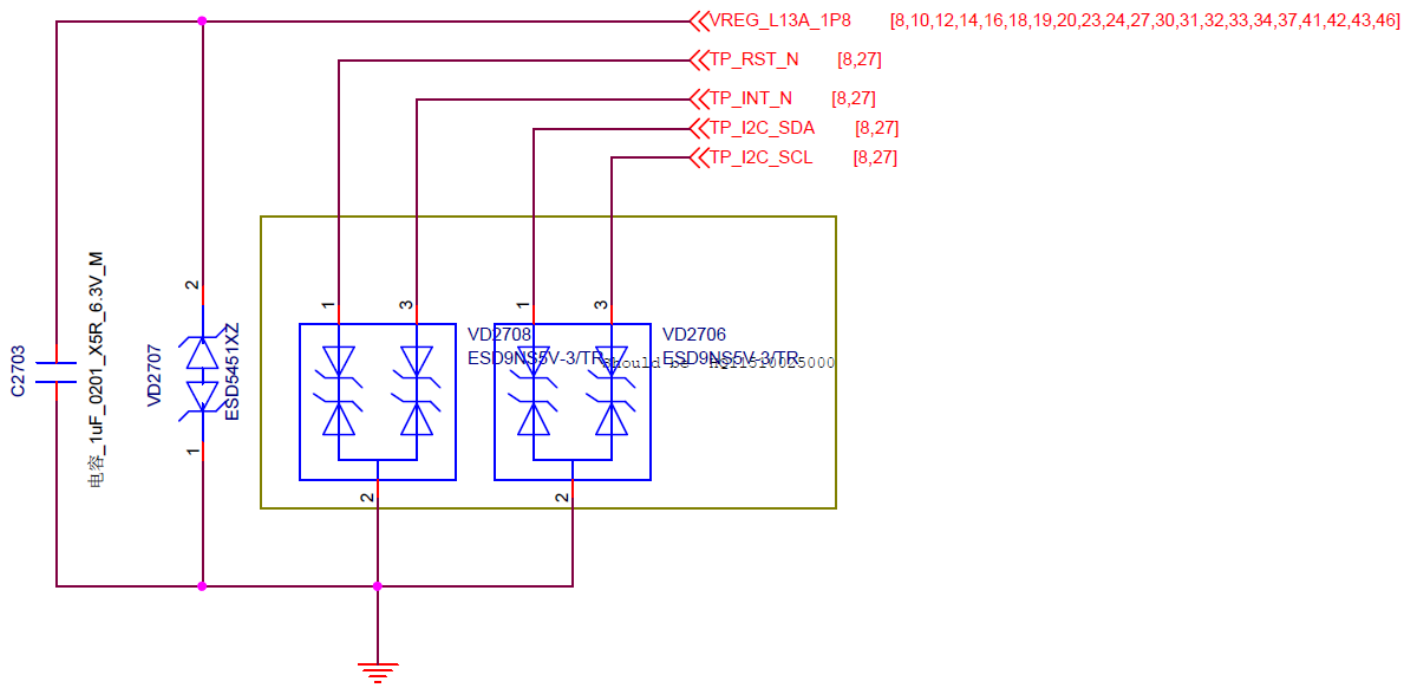


- 1 Re-install LCD display screen or replace it before powering on the phone and check again;
- 2 Check LCD connector or its soldering for errors;
- 3 Check LCD circuit resistors (R2701, R2704, R2708, R2721, R2720) for soldering errors or damage;
- 4 Check CPU for over-gluing, in case an X-ray equipment is available, check the IC solder ball with it;
- 5 Add-solder or re-solder CPU or replace it;





## 5. Touch control screen errors

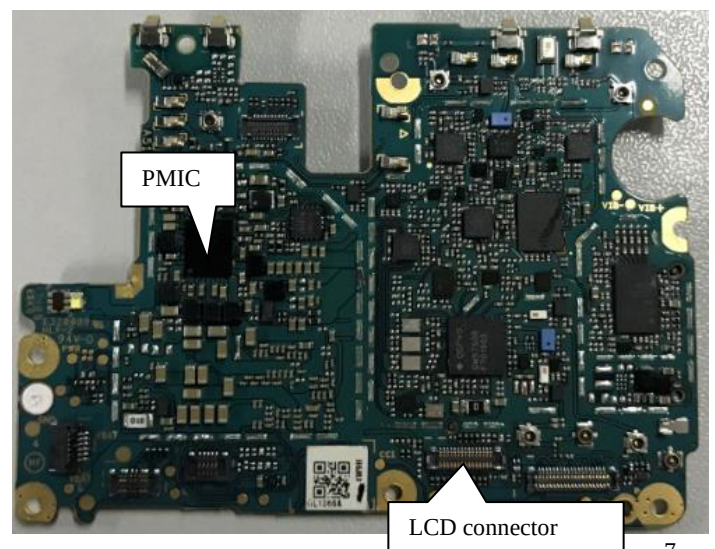


### Solution:

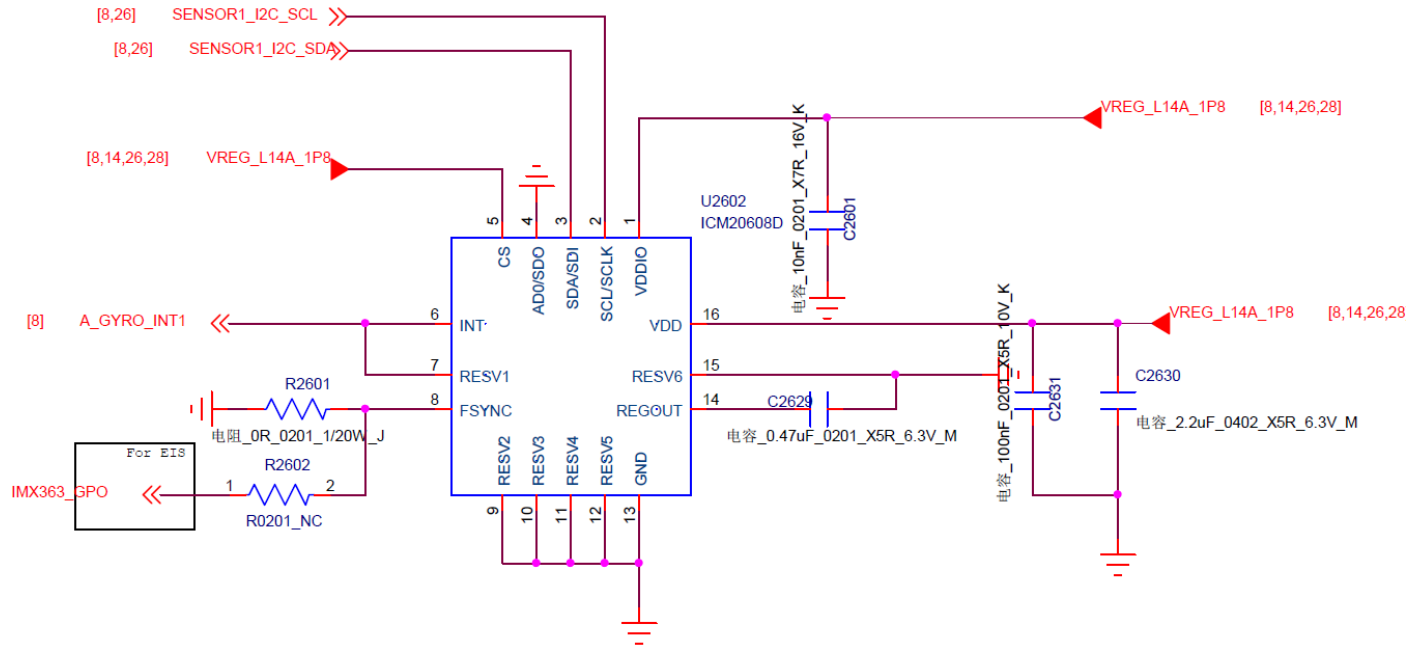
- 1 Re-install TP or replace it before powering on and testing it again;
- 2 Check LCD connector J2701 or its soldering for any errors, re-solder or replace it before testing it again;
- 3 Check the CPU soldering status with an X-ray, add-solder or re-solder it before testing it again;
- 4 Measure TP test signal point's resistance to ground with a multimeter, compare it against a normal board;
- 5 Replace CPU;



- 1 Check optical chip surface for dirt or blockage;
- 2 Check optical/distance sensor for damage or soldering error;
- 3 Measure VREG\_L14A\_1P8 voltage;
- 4 Measure whether the VREG\_L3B\_3P0 voltage is normal;
- 5 Add-solder or re-solder CPU/PMIC or replace it;
- 6

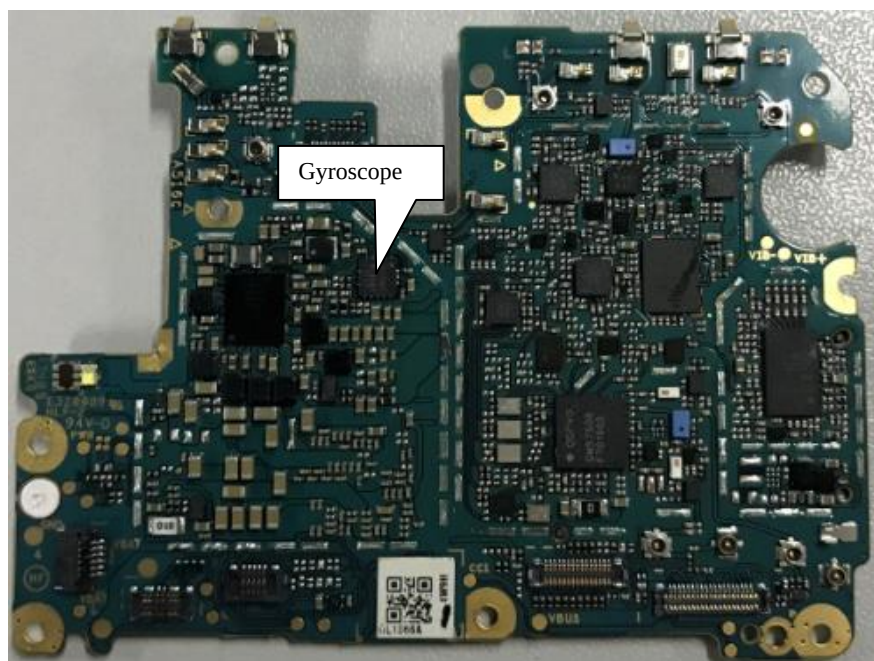


## 7. G-sensor failure



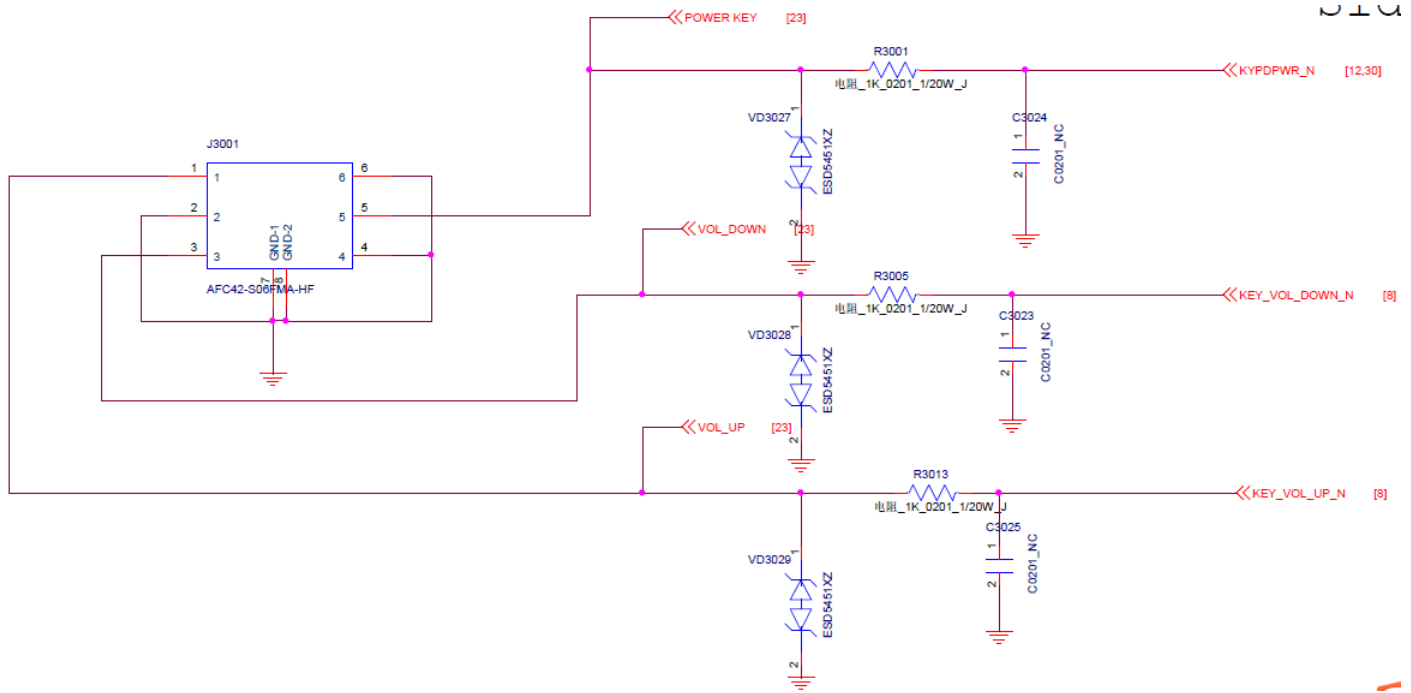
### Solution:

- 1 Measure whether the VREG\_L14A\_1P8 voltage is 1.8V
- 2 Re-solder or replace gravity sensor U2602;
- 3 Check gravity sensor module circuit components for soldering errors or damage;
- 4 Re-soldering or replace CPU;



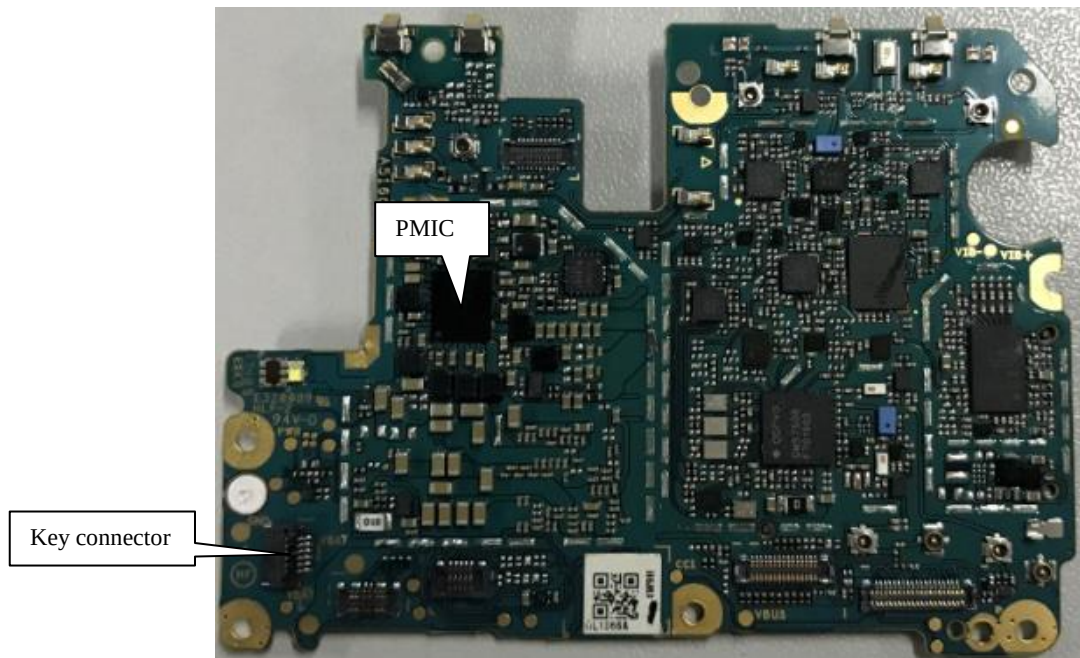


## 8. Key failure

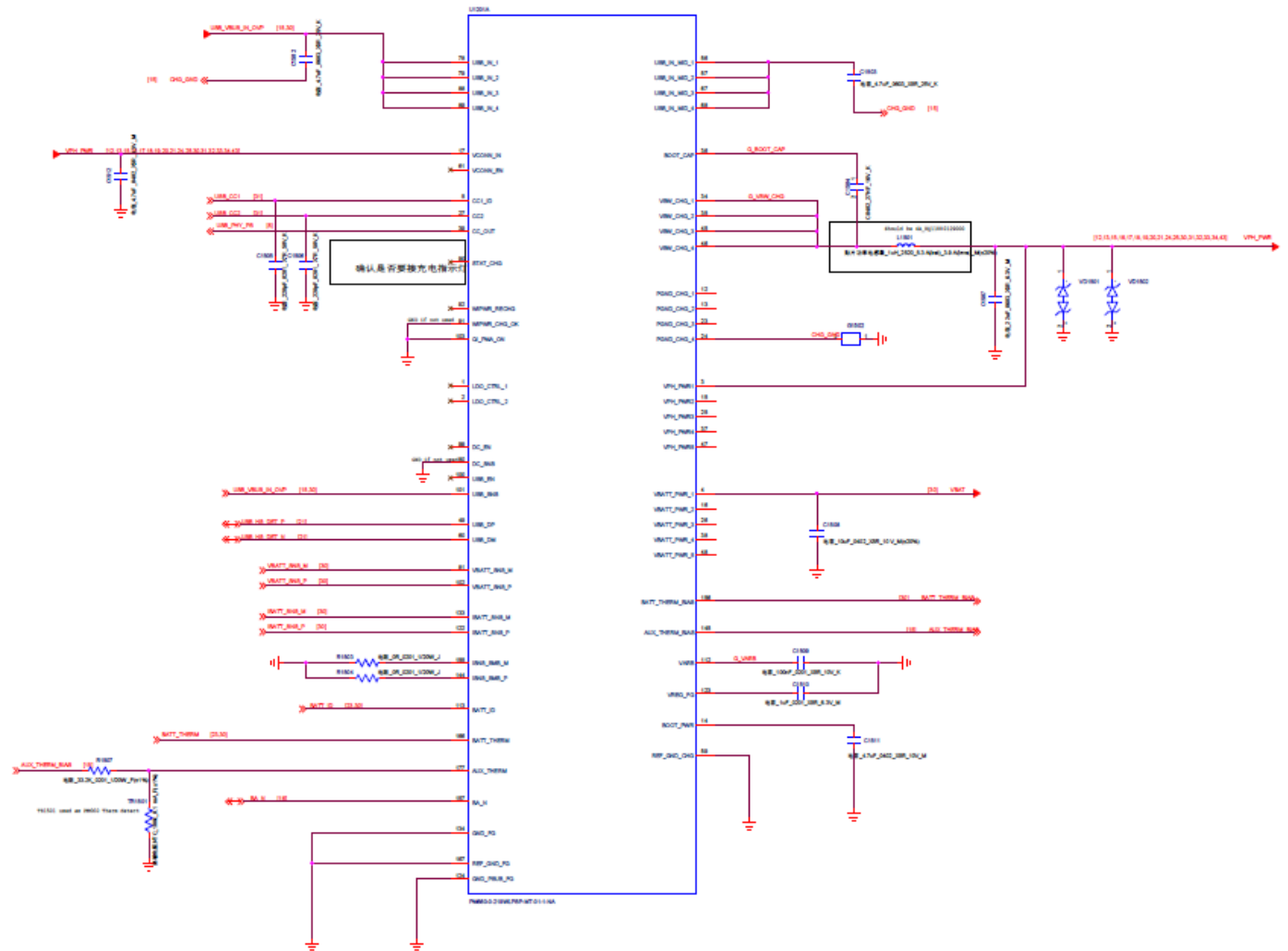


### Solution:

- 1 Re-install connector or replace side key before testing again;
- 2 Check whether POWER KEY/KYPD PWR\_N voltage signal is normal;
- 3 Check components on key circuit for damage or soldering errors;
- 4 Re-solder or replace the CPU or PMIC;

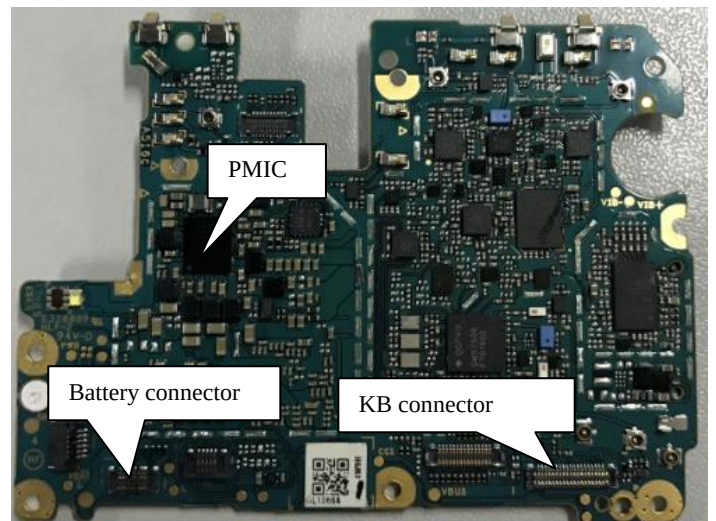
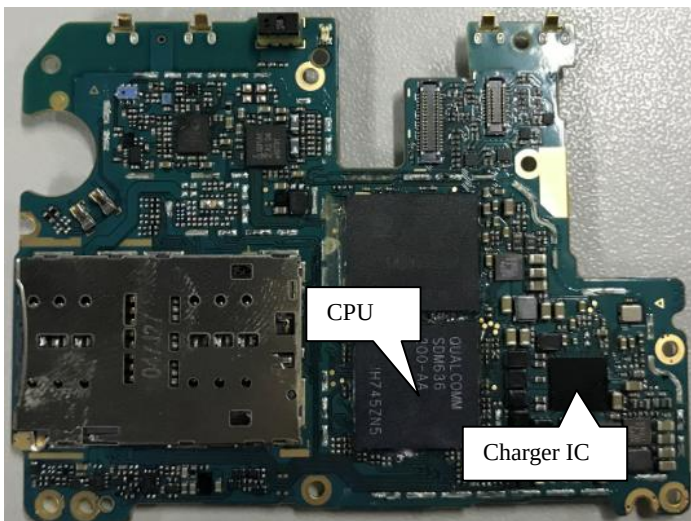


## 9. Charging failure, USB identification failure



### Solution:

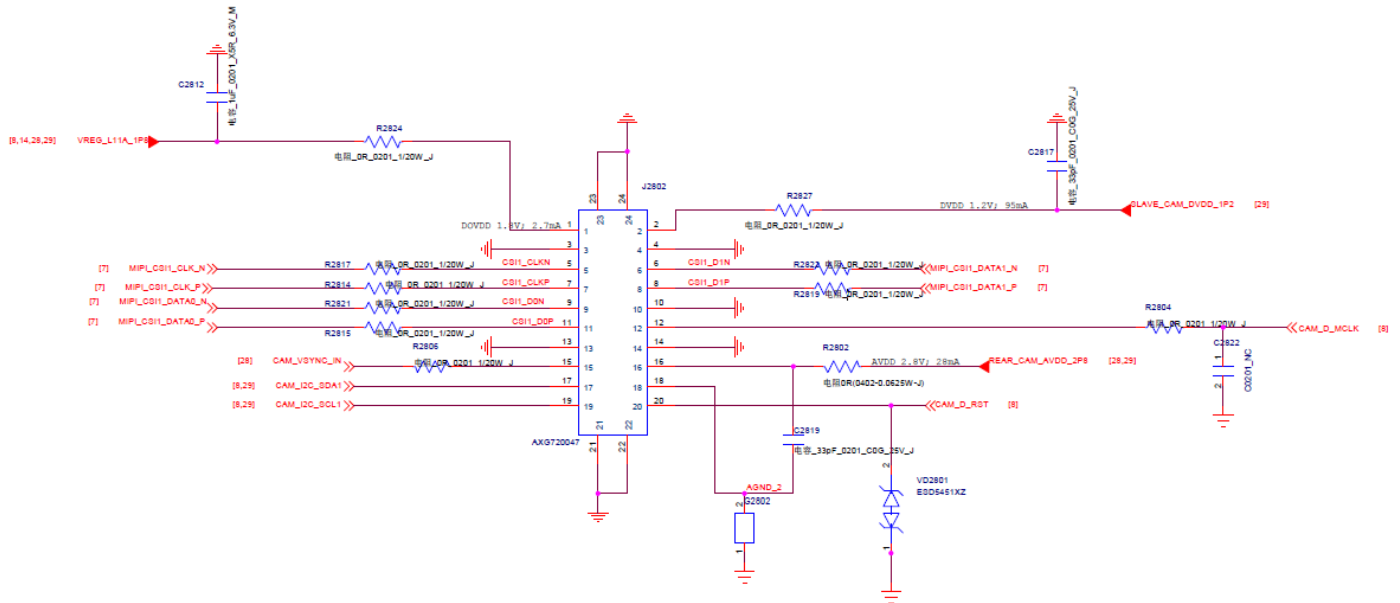
- 1 Re-connect KB board or replace KB board before testing it;
- 2 Check MB's KB board connector for dirt or damage;
- 3 Replace battery before testing again;
- 4 Check battery connector J3001 for dirt, bubble soldering, damage;
- 5 Check charging management chip U1201 for soldering error;
- 6 Re-solder U1601 power management chip and check for soldering errors;
- 7 Re-solder or replace the CPU





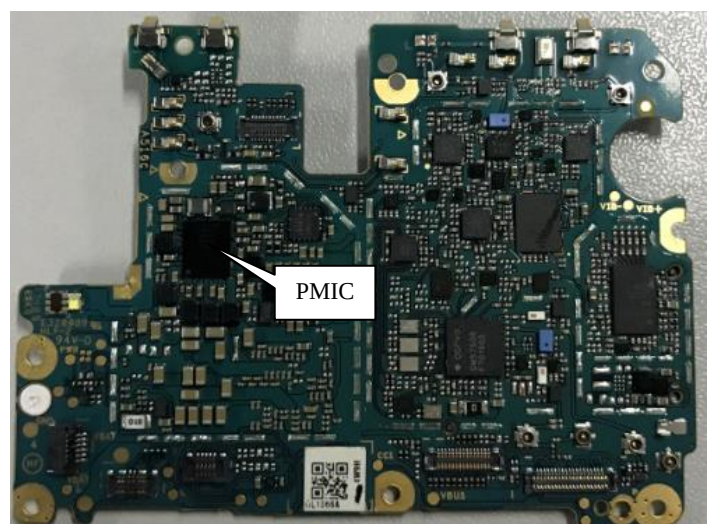
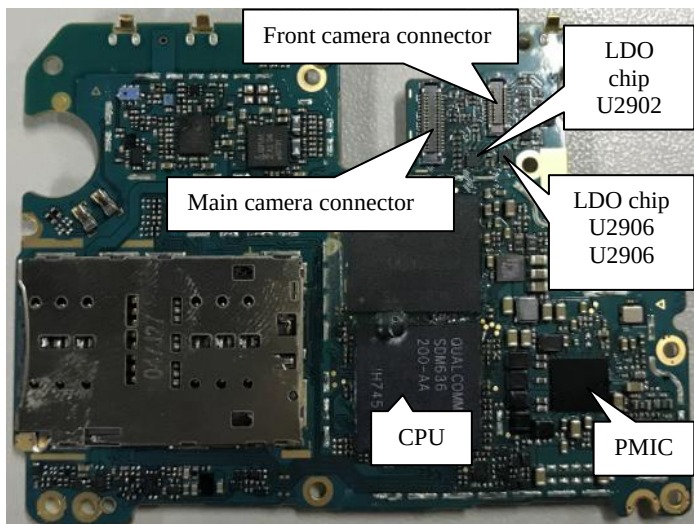
- 1 Check camera installation for any abnormalities, re-install it or replace it before testing again;
- 2 Check the connector of the main camera J2801 and the peripheral resistors (R2823/R2826/R2829) for damage or soldering errors, re-solder or replace the connector;
- 3 Check VREG\_L11A\_1P8 /VREG\_L8B\_2P8 voltage;
- 4 Check LDO chip U2902 for damage or soldering errors;
- 5 Add-solder or re-solder PMIC or replace it;
- 6 Add-solder or re-solder CPU or replace it;

## 11. Auxiliary camera function failure



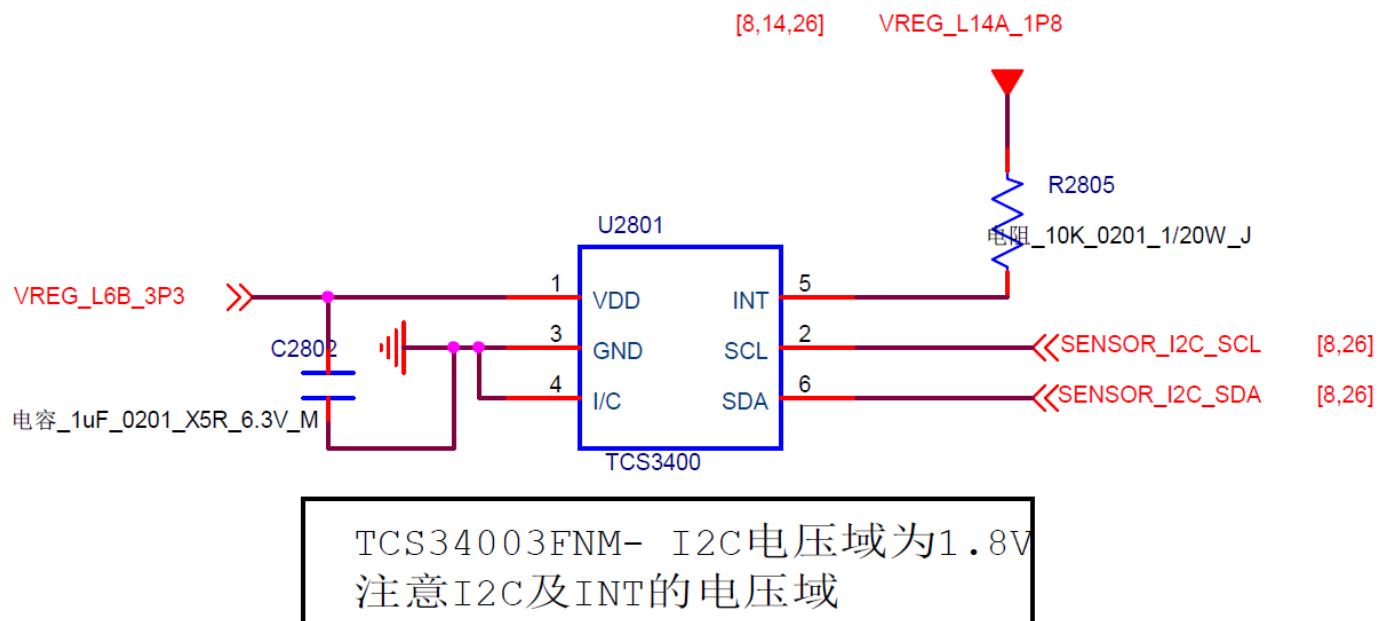
### Solution:

- 1 Check camera installation for any abnormalities, re-install it or replace it before testing again;
- 2 Check the auxiliary camera J2802 connector for damage or soldering errors, re-solder or replace the connector;
- 3 Check VREG\_L11A\_1P8 /VREG\_L8B\_2P8 for voltage errors, if so, check PMIC for output errors or peripheral circuits for short circuits or broken circuits;
- 4 Check whether the LDO chip U2906 or its soldering is normal;
- 5 Add-solder or re-solder CPU or replace it;



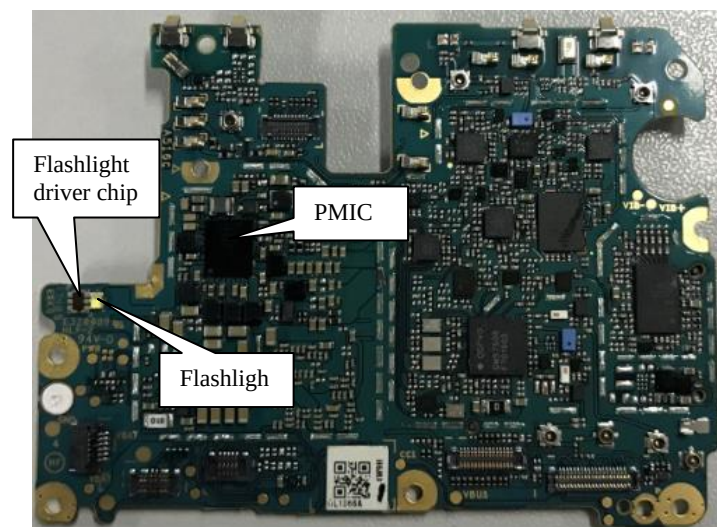


## 12. Flashlight failure

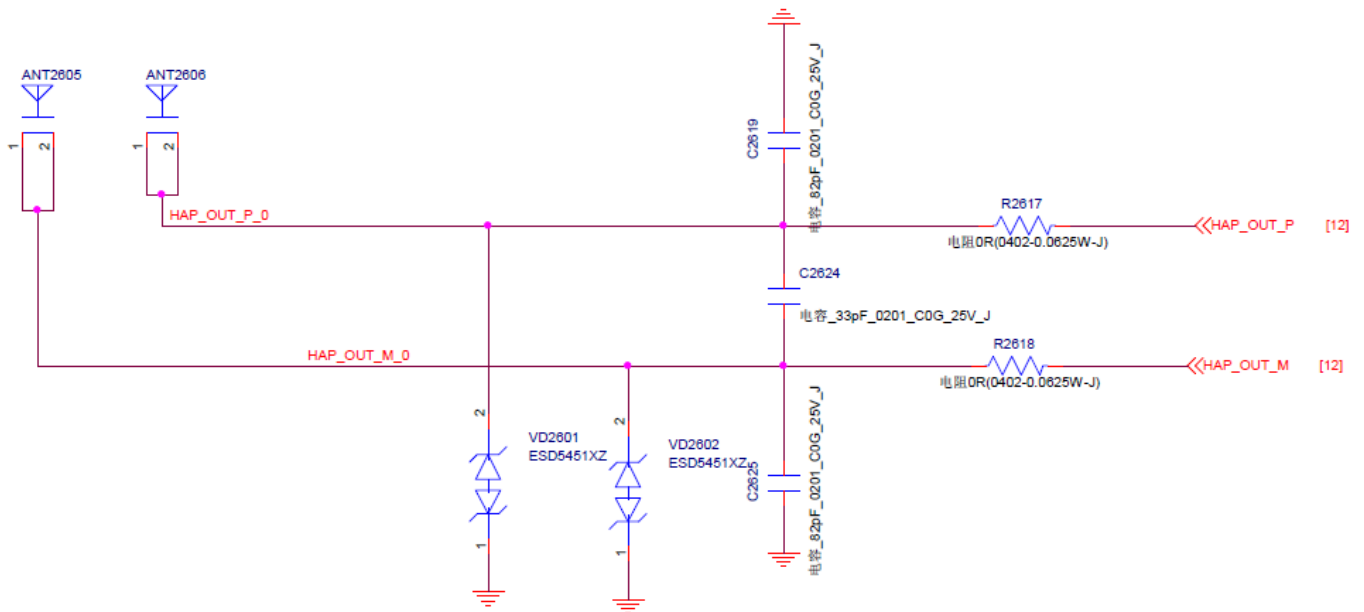


### Solution:

- 1 Add some 3.0V voltage to soldering tray at both ends of LED to test it for lighting up;
- 2 Add-solder or replace flashlight D2603;
- 3 Re-solder or replace flashlight drive U2801;
- 4 Re-solder or replace PMIC U1601

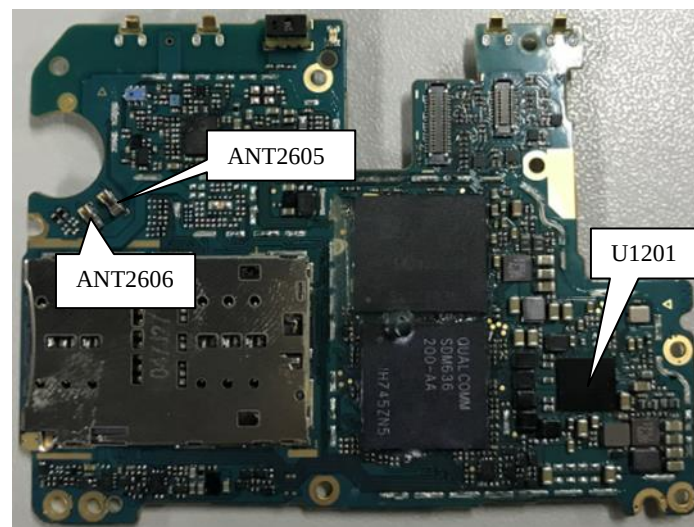


## 13. Motor error

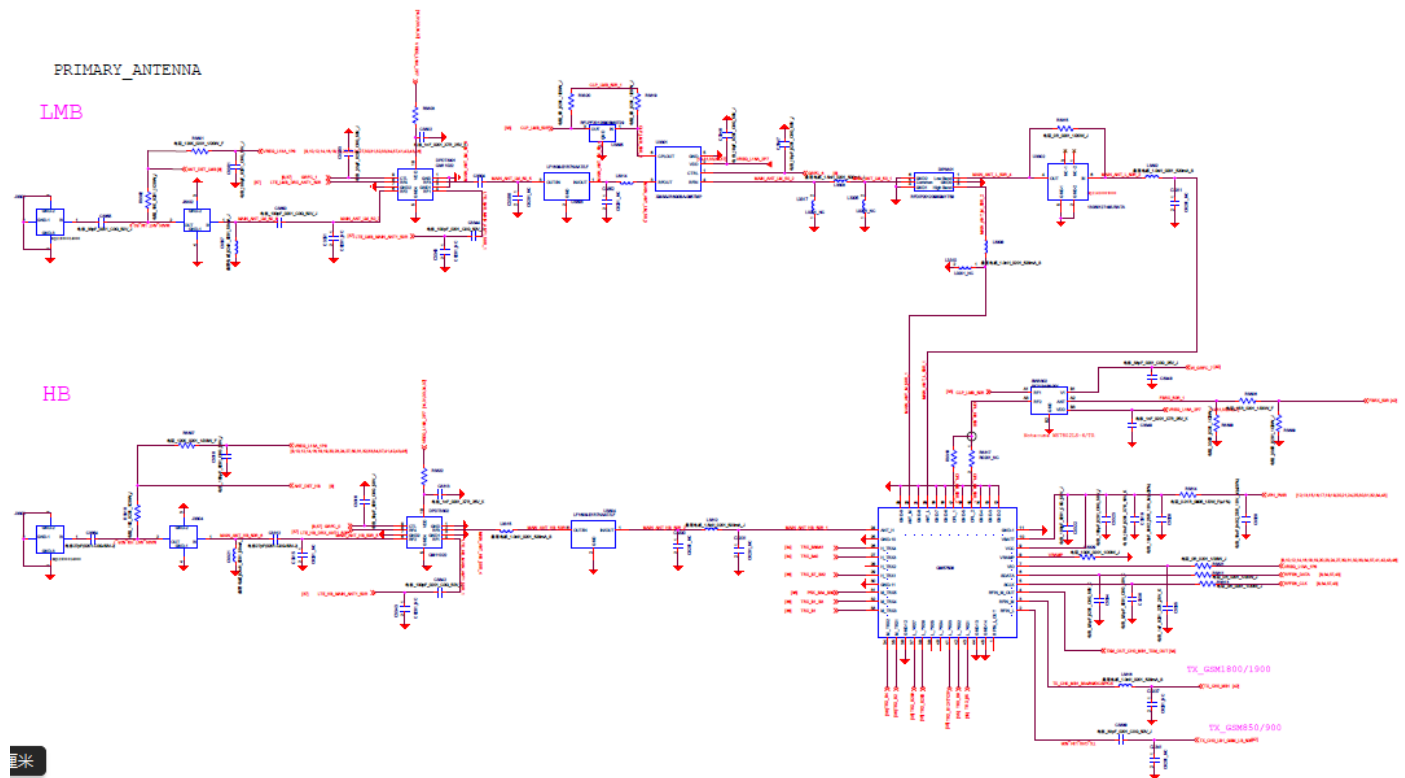


### Solution:

- 1 Measure the voltage on the ANT2605 and ANT2606 shrapnel of the KB board; compare against normal MB;
- 2 Re-solder motor or replace before testing it again;
- 3 Re-solder power management chip U1201 or replace it;



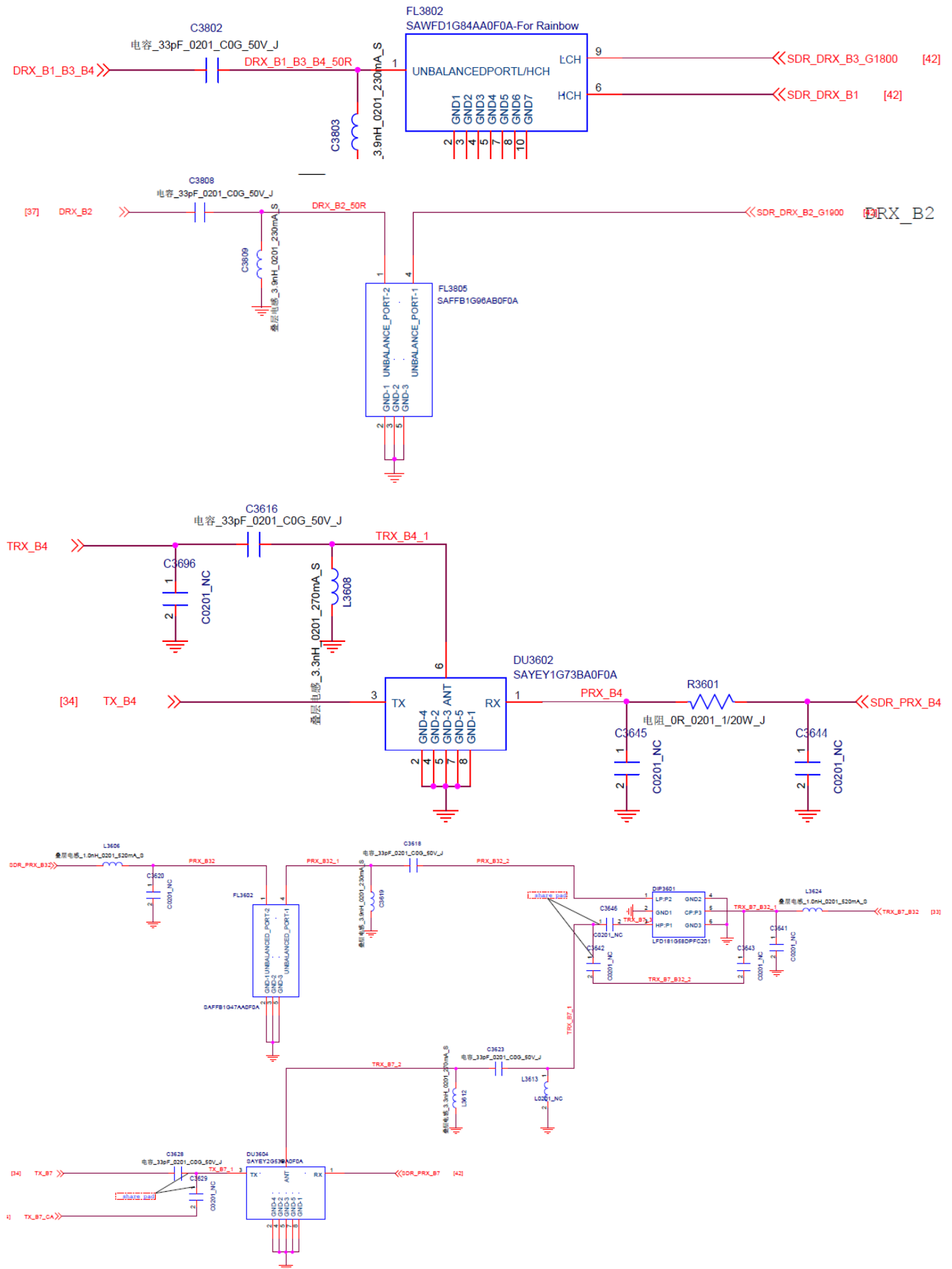
## 14. GSM network failure



### Solution:

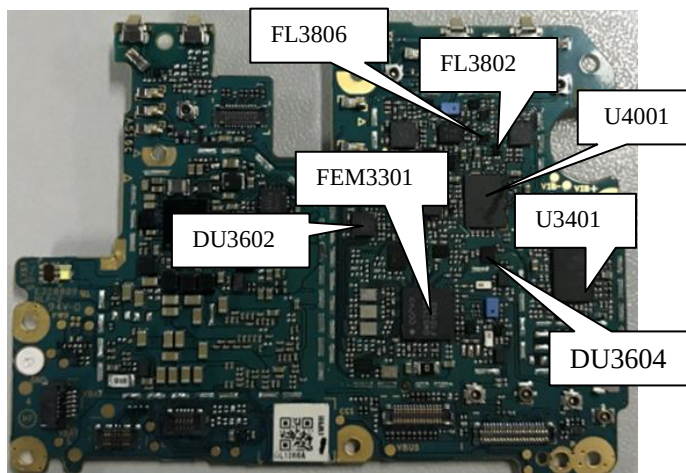
- 1 Check KB board's antenna soldering status, contact with phone casing antenna, or damaged body;
- 2 Check RF connector connection circuit induction resistance components for error soldering after damaged;
- 3 Check the RF PA chip FEM3301 and the relevant components and its soldering for damage;
- 4 Add-solder or replace the U4001 RF transceiver chip;
- 5 Add-solder or replace U1601 power management chip;
- 6 Add-solder or replace U501 main chip;

## 15. WCDMA & LTE network failure

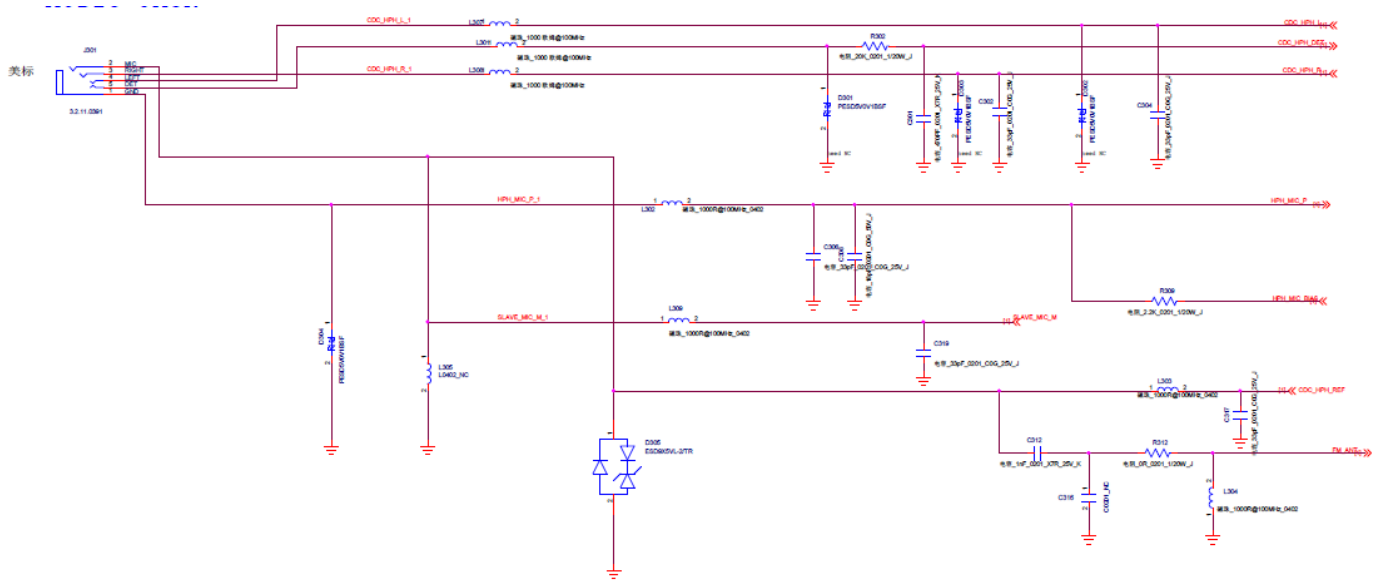




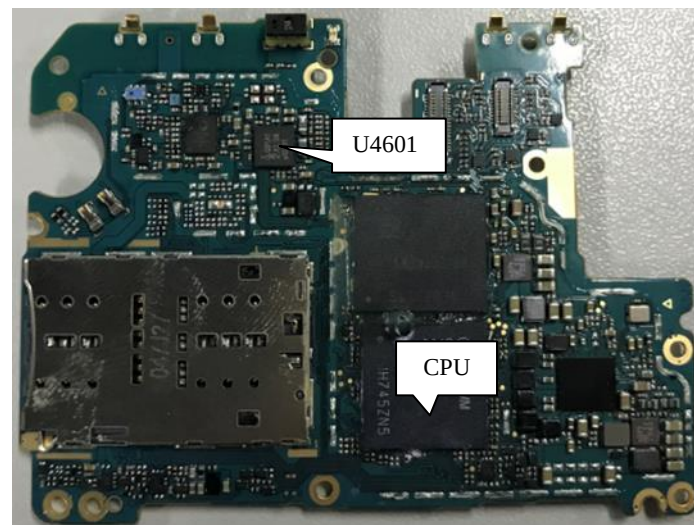
- 1 Band1 transceiver failure, re-solder U3401, FEM3301, filter FL3802, RF transceiver U4001;
- 2 Band2 transceiver failure, re-solder U3401, FEM3301, filter FL3805, RF transceiver U4001;
- 3 Band4 transceiver failure, re-solder U3401, FEM3301, filter DU3602, RF transceiver U4001;
- 4 Band7/Band32 transceiver failure, re-solder U3401, FEM3301, filter DU3604, FL3604, RF transceiver U4001;
- 5 Band8 transceiver failure, re-solder U3401, FEM3301, filter FL3903, RF transceiver U4001;
- 6 Band34/39 transceiver failure, re-solder or replace filter FL3806, RF power amplifier U4202, RF transceiver U4001;
- 7 According to RF indicator, check relevant band's transceiver channel components and their soldering for damage;
- 8 Re-solder or replace U501 CPU;







- 1 Re-test FM function after replacing headset;
- 2 Check headset jack for damage, its soldering for skew or over height;
- 3 Add-solder or re-solder the U4601 WLAN/BT/FM/GPS 4-in-1 chip, or replace it before testing again;
- 4 Add-solder or re-solder CPU U501 or replace it before testing it again;



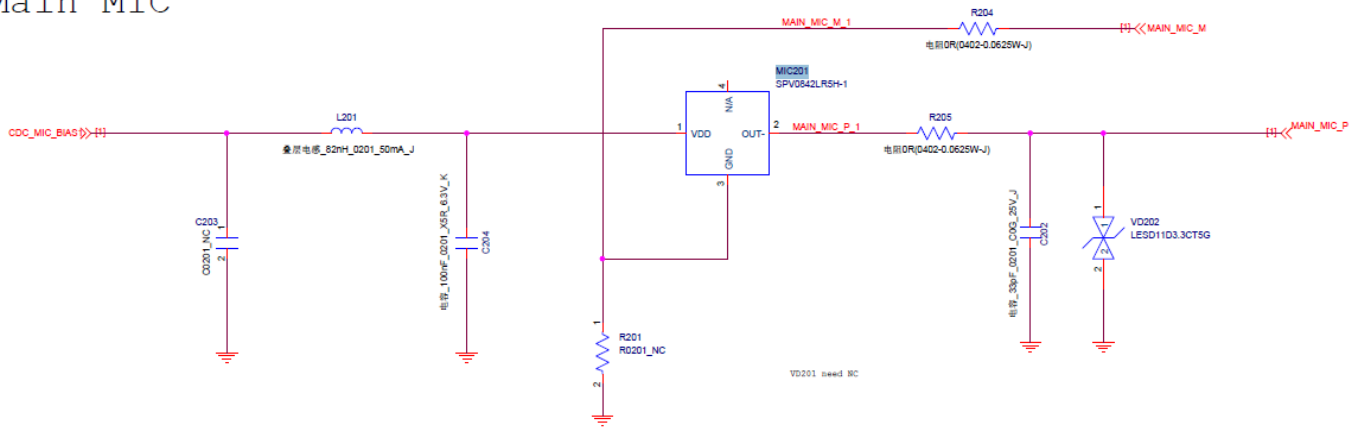


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- A photograph of a green printed circuit board (PCB) with various electronic components. A white speech bubble points to a large, dark, rectangular integrated circuit (IC) labeled 'CPU'. The board also features a large black component on the left, numerous smaller chips, and various connectors and solder points.

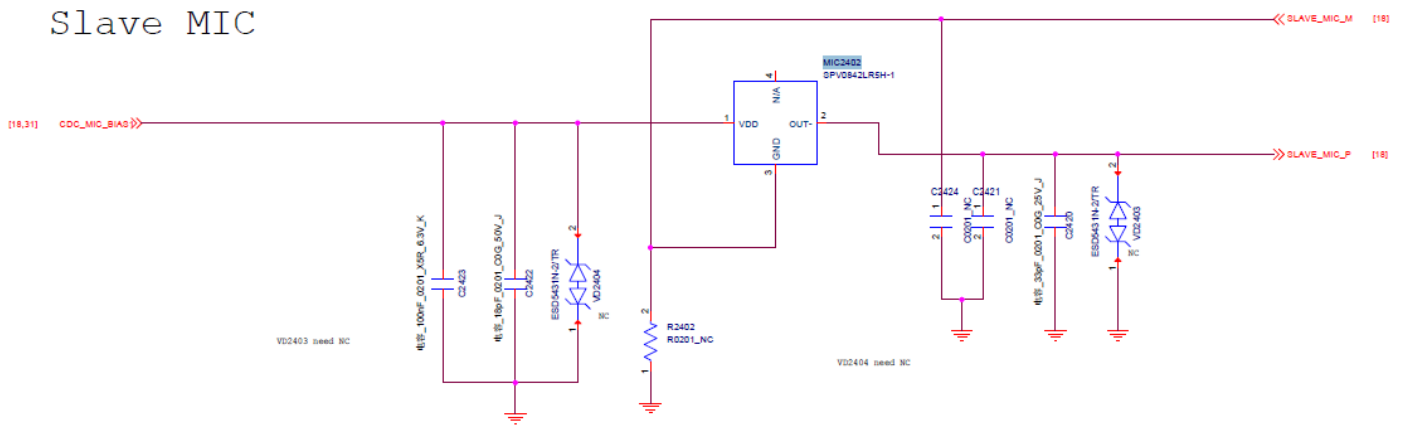


## 19. MIC silent/low volume

### Main MIC

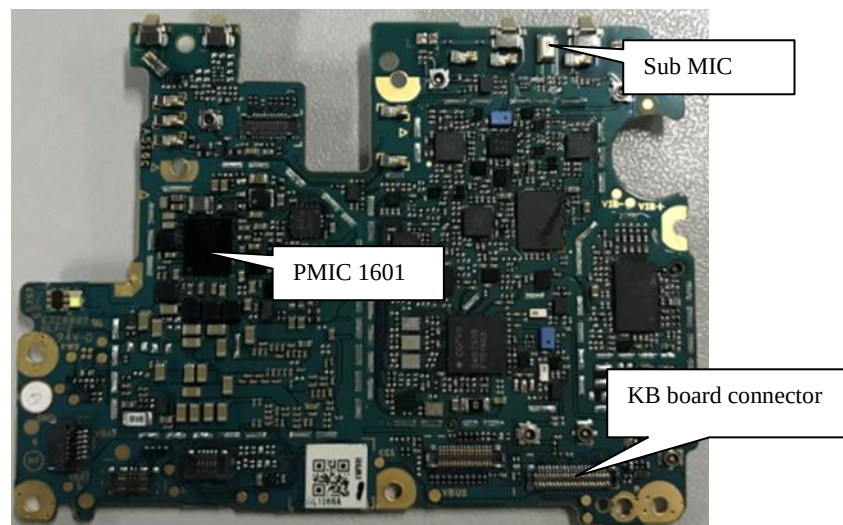


### Slave MIC



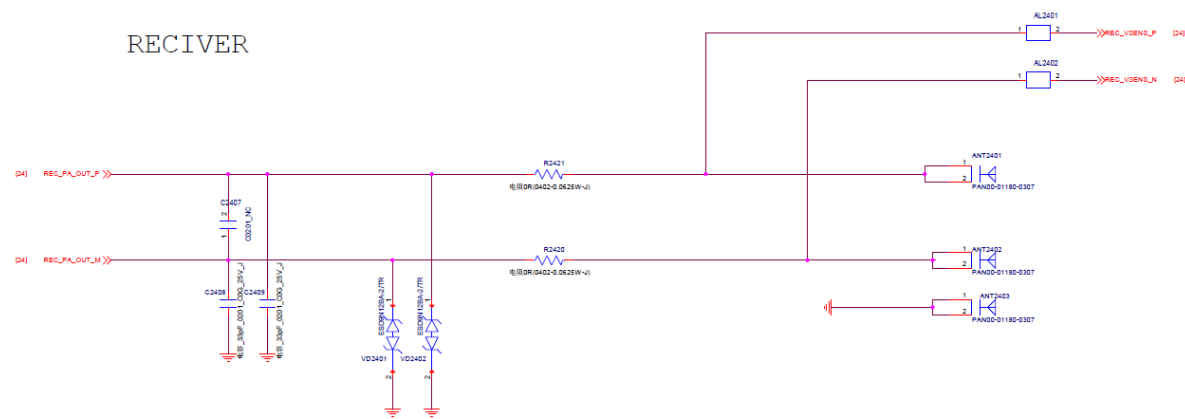
### Solution:

- 1 Main MIC is located on the KB board, replace KB board or MB positioning point is on MB or small board;
- 2 Check MB's KB board connector for damage or soldering for skew or continuous tin;
- 3 Visually check secondary MIC relevant components for soldering offset or body damage;
- 4 Replace secondary MIC 2402 before testing it again. Pay attention to high temperature during replacement as it may damage it;
- 5 Add-solder or re-solder U1601 PMIC, or replace it before testing again;





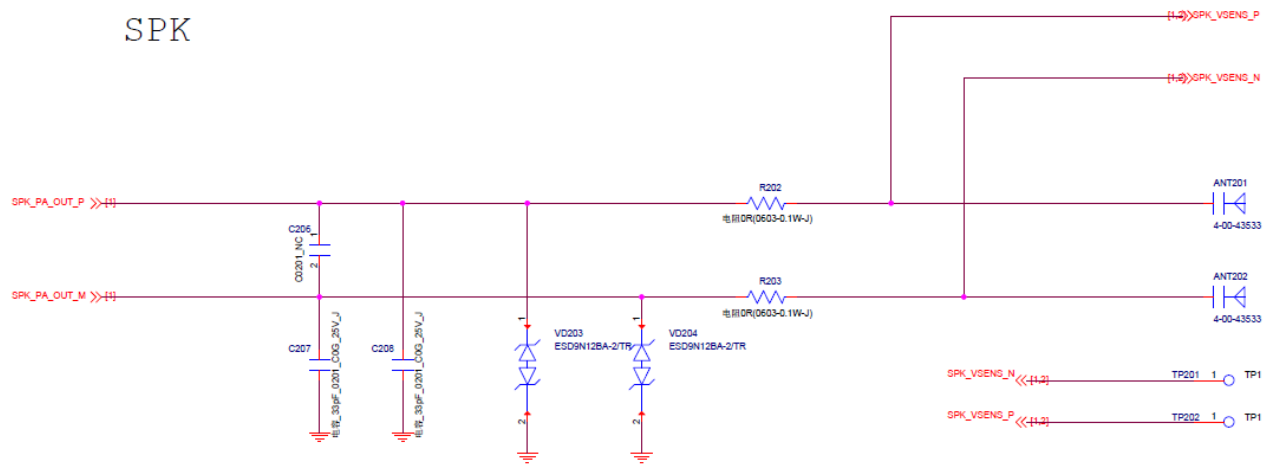
## 20. Earpiece silent



### Solution:

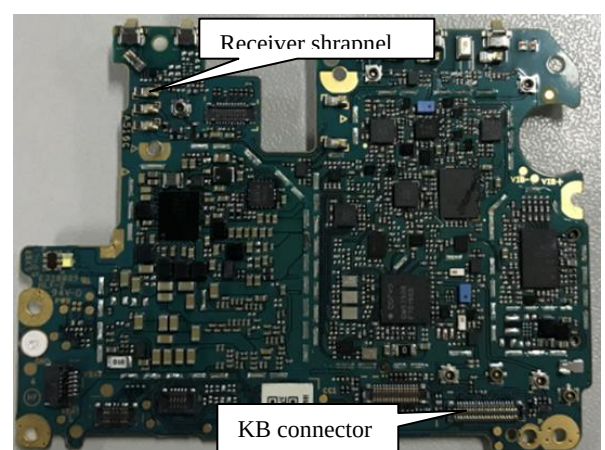
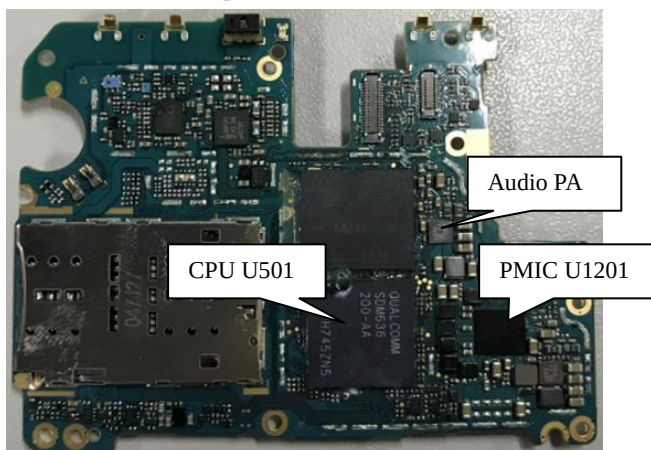
- 1 Re-install or replace earpiece before testing it again;
- 2 Visually check headset circuit components for soldering offset or body damage;
- 3 Add-solder / re-solder the audio smart amplifier U2402 before testing it again.
- 4 Add-solder or replace PMIC U1201

## 21. Speaker silent



### Solution:

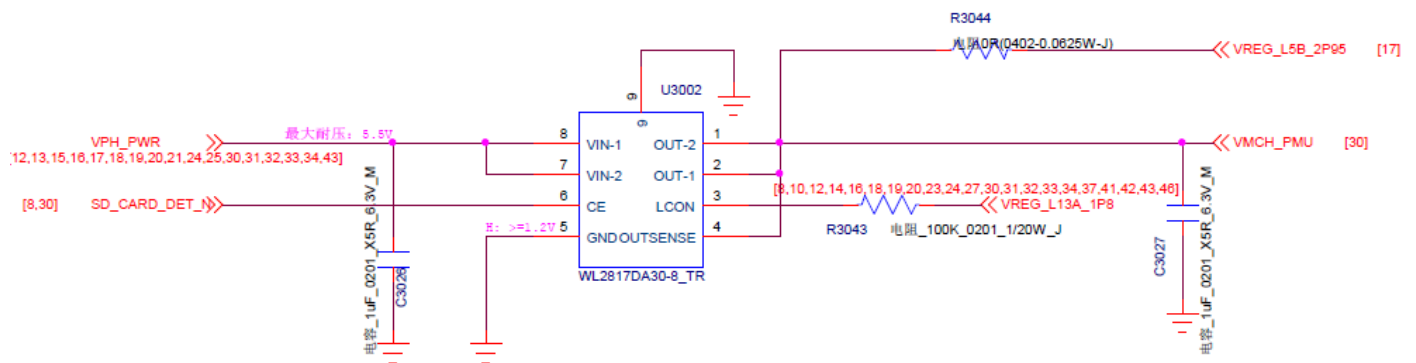
- 1 The speaker is located on KB board. Replace KB board to test and determine whether KB board or the MB is defective and then repair it;
- 2 Visual check relevant components for damage or soldering error;
- 3 Check connector between MB and KB board for skew and continuous time;
- 4 Add-solder or re-solder U2402/ U2401 Audio PA or replace it;
- 5 Add-solder or replace PMIC U1201.





- 1 Re-install SIM card or replace before testing it again;
- 2 Visual check SIM tray for damage, soldering for skew or overweight;
- 3 Measure the SIM power supply voltage VREG\_L17A\_UIM2、VREG\_L15A\_UIM1;
- 4 Re-solder or replace U1201 PMIC;
- 5 Re-solder or replace U501 CPU;

## 23. T-flash card identification failure



### Solution:

- 1 Re-install T-flash or replace it before testing again;
- 2 Visually check the T card tray for damage, soldering for skew or over height;
- 3 Measure T-flash power supply voltage VPH\_PWR;
- 4 Re-solder or replace U1201 PMIC;
- 5 Re-solder or replace U501 CPU;

